Supporting Information: Basin hopping on Lennard-Jones clusters with one preset at every size

Rohit Goswami,[†,‡,¶] Ruhila Goswami,[§] Amrita Goswami,[‖] Moritz Sallermann,[‖] Sonaly Goswami,[⊥] and Debabrata Goswami[*,#,⊥]

†*SURF B.V., Amsterdam*

‡*Science Institute, University of Iceland, Reykjavik, Iceland*

¶*TurtleTech ehf., Reykjavik, Iceland*

§*Faculty of Life and Environmental Sciences, University of Iceland, Reykjavik, Iceland*

‖*Universidad Complutense de Madrid, Madrid, Spain*

⊥*Department of Chemistry, Indian Institute of Technology Kanpur, Kanpur, India*

#*Center for Bits and Atoms, Massachusetts Institute of Technology, Cambridge, Massachusetts, United States*

E-mail: dgoswami@iitk.ac.in

1 Board constants

We read a hit as an energy strictly below the reference plus 0.001, taken from a converged improvement. The references, in reduced units, are -173.928427 for LJ_{38}, -279.248470 for LJ_{55}, and -397.492331 for LJ_{75}. LJ_{98} uses -543.665361, and LJ_{104} uses -582.086642. The temperature is 0.8, the value Wales and Doye used for. Lennard-Jones basin hopping and the value stored at `preset.rs:8`.[S1] The start density passed to `random_cluster` is 0.7. The

function comment calls the draw liquid-like.

2 Plain walk

We run one further process on the recommended preset at 38 atoms. With the stall exits. And the Bayes screen turned off, seed 0, budget 30000. We keep the record in `logs/lj38-plain-seed0.log`. The best energy in that log is -173.928427. After 30000 charged evaluations.

3 Published symmetrised costs

`data/literature.tsv` quotes Oakley, Johnston and Wales, Table 1.[S2] The continuous-symmetry basin-hopping mean. At 38 atoms is 4369 evaluations over 100 successful runs. The same file's Table 2 rows are genetic-algorithm means. At 75 and 98 atoms. No row in that file is a genetic algorithm at 104 atoms.

4 TIP4P pair constants

`logs/tip4p_constants.txt` records Jorgensen's $A = 2510400$ and $C = 2552.24$.[S3] The units are kJ Å^{12} per mole and kJ Å^{6} per mole. The Wales-group basin-hopping program (GMIN) matches those two numbers. In C_6 and C_{12}. A rounded well depth and a rounded σ give the anneal pair. $C_6 = 2551.8252$ and $C_{12} = 2510336.2$. These two coefficients are the TIP4P pair in the log. The GFN2-xTB hexamer rows are a separate comparison.

5 Earlier campaign

`data/campaign-scores.csv` is an earlier 64-seed campaign, and on $TIP4P_{20}$ a 16-seed campaign. Its recommended rows were scored. With a preset that still branched on the atom

count. `data/lj55_lj98/seeds.csv` scores LJ_{98}. By the energy rule. The raw run logs for the LJ_{98} rows of that seed file are not in this record. None of those rows is the LJ_{75} campaign of 200 seeds. OGOLEM, cited as Dieterich and Hartke, is in the campaign file at 6 of 64 on LJ_{104}.[S4] ABCluster, cited as Zhang and Dolg. Is in the same file.[S5] The seed file is the record of those LJ_{98} medians.

6 Hardware and reproduction

We keep the LJ75 receipt in `logs/lj75-hardware-receipt.txt`. We run `scripts/lj75_hardware_capture.sh`. That script writes the file. From the live process. The encounter-curve binary was built from tree. 37535ec1a6e99d47a2afe0752072a33af64dccb3. The tree of the unpublished build commit. A public reader gets the archived blob-hash check only. That tree has copies of `bias.rs`, `neighbors.rs` and `potentials.rs`, the hot-path bench file, the newsfragment. and `precision-policy.org`. It lacks the `cluster_hopping.rs` and `tip4p.rs` parts of that copy and the bench entry in `Cargo.toml`. Its `rigid_body.rs` is a third version. Main's sealed LJ75 counts used restart on stall. And an empty surface list. This article uses the preset with restart left off. The encounter-curve figure draws that earlier campaign: the recommended. Hop, the derived hop, and the Wales-group basin-hopping program (GMIN), on 200 seeds. The panel prints that run's PAR-2 values. We keep the source archive at `logs/lj75-source-37535ec1.tar.xz`. It includes `LICENSE.txt`. The notice reads Copyright (c) 2023 anneal core developers. `scripts/lj75_rebuild_from_archive.sh` rebuilds the example. With the bank-rpc feature. We keep the build receipt in `logs/cargo-build-receipt.txt`. Against commit `da65cc6d`, `cluster_hopping.rs` differs by 218 insertions and 2 deletions, `tip4p.rs` by 93 and 15, `rigid_body.rs` by 52 and 14, `examples/molecular_cluster.rs` by 1 insertion, `Cargo.toml` by 4 insertions and 1 deletion, and `Cargo.lock` by 55 deletions. On the recommended LJ_{75} path the hop reuses a pure. Distance descriptor. The added tests require a second call on the same coordinates to

match. Bit for bit. The smooth overlap of atomic positions (SOAP) key still recomputes, because its basis learns. The hexamer engine digest is the `engine_sha256` line of. `water/VERSIONS` inside `logs/logs-water.tar.xz`. That digest is. 4836648ee77203e9462d51e25 a1970b3f0df3a3bcd126f9a7454f6896ac66df8. The hexamer wall-time medians are in `logs/hexamer-walls.txt`.

7 Structure plates, sketch-maps, and landfold

Reference plates: S1 and S2. Sketch-map of LJ_{38}: S3. Landscape of that size: S4. LJ_{75} panel: S5. LJ_{98} panel: S6. Fold plots: S7, S8, S9, S10.

8 Copper slab

The hit tolerance is 0.001 eV. The script is `scripts/elja_hq_sci_molslab_bank.sh`. The lines are 118 and 119. TARGET_TOL there is 1e-3. The reference is -415.971529 eV. The per-seed gaps run from 8e-6 eV to 3.68e-4 eV above that reference. Each gap misses the reference energy. Eight seeds lie inside the 0.001 eV tolerance.

References

(S1) Wales, D. J.; Doye, J. P. K. Global optimization by basin-hopping and the lowest energy structures of Lennard-Jones clusters containing up to 110 atoms. *J. Phys. Chem. A* **1997**, *101*, 5111–5116.

(S2) Oakley, M. T.; Johnston, R. L.; Wales, D. J. Symmetrisation schemes for global optimisation of atomic clusters. *Phys. Chem. Chem. Phys.* **2013**, *15*, 3965–3976.

(S3) Jorgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. L.

(a)

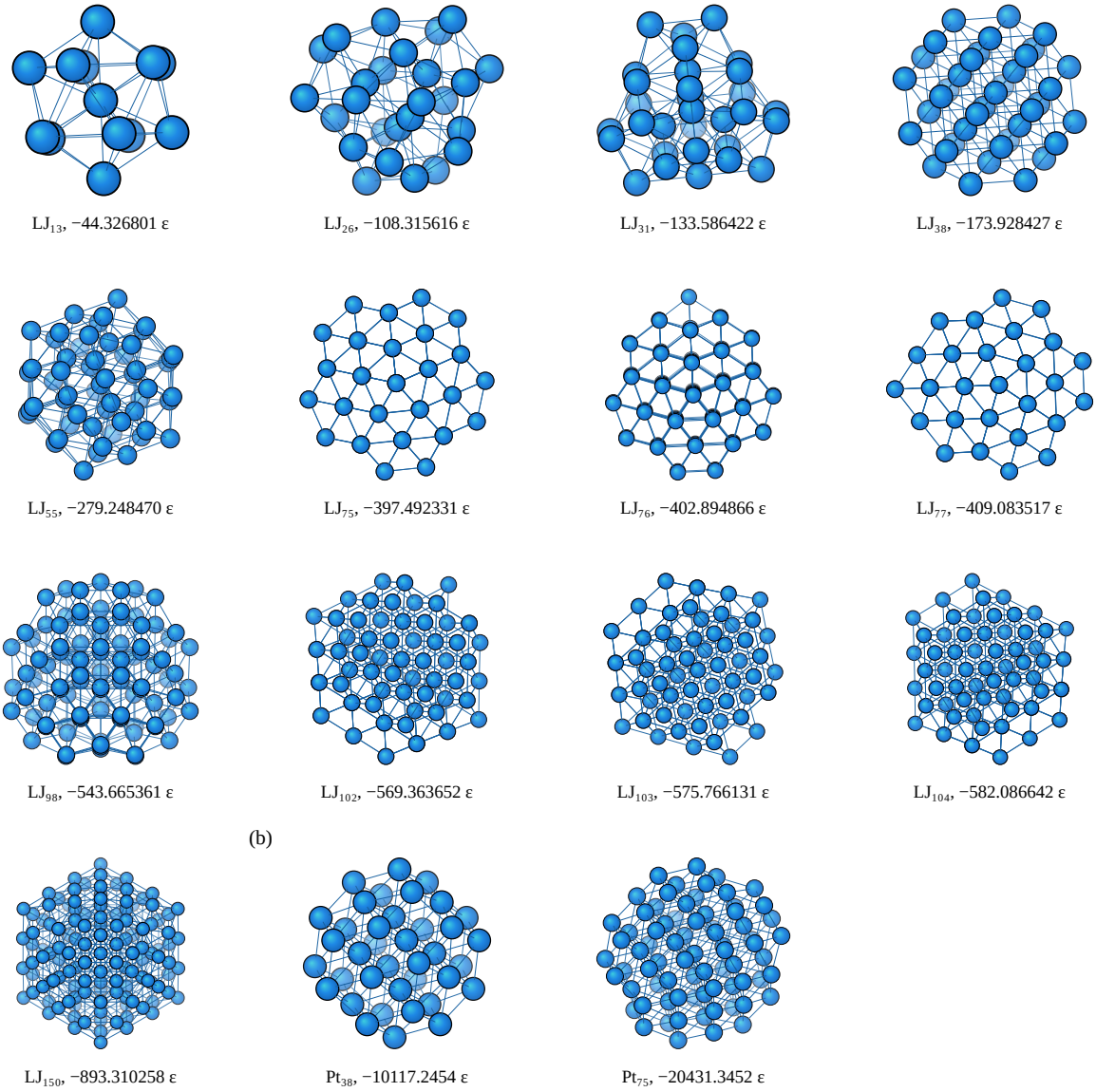


Figure S1: Reference minima of the benchmark.

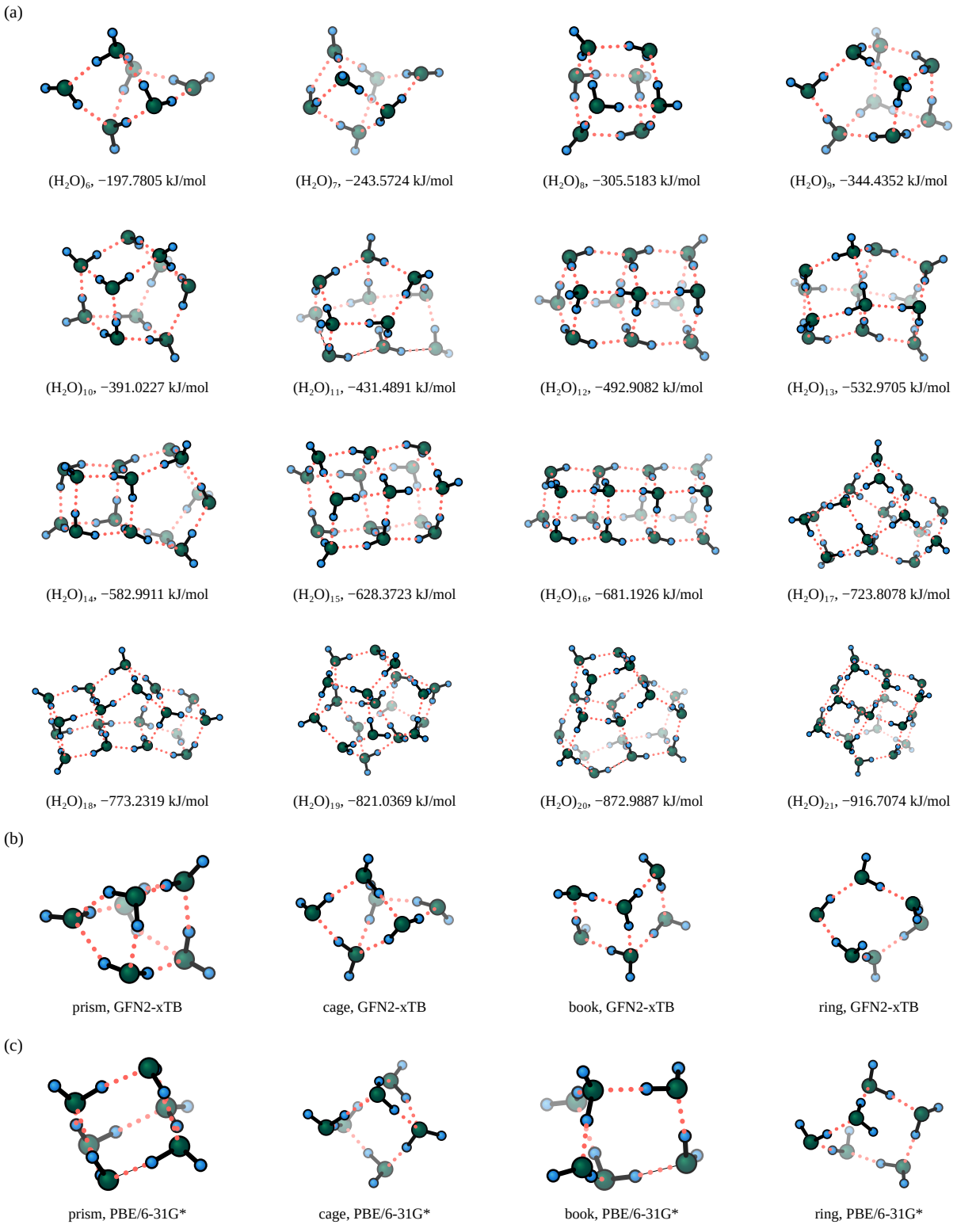


Figure S2: TIP4P plates and the hexamer classes. The hexamer plate prints −197.7805 kJ/mol. That is the Cambridge Cluster Database value. It uses the Jorgensen TIP4P constants C6 = 2552.24 and C12. = 2510400. anneal uses $\varepsilon = 0.6485$ kJ/mol and. $\sigma = 3.15365$ Å. Its C6 is 2551.8252 and its C12 is 2510336.2.

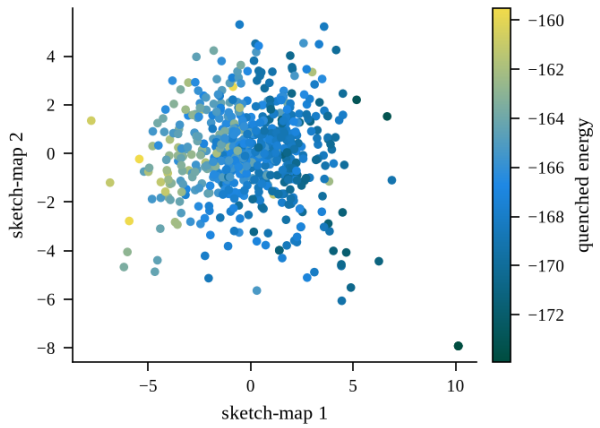


Figure S3: LJ_{38} sketch-map of the sealed landmarks. Colour is the quenched energy. The 600 landmarks pool basin hopping, the recommended hop. Random structure search and two references. The darkest point is the fcc minimum, which the recommended hop reaches.

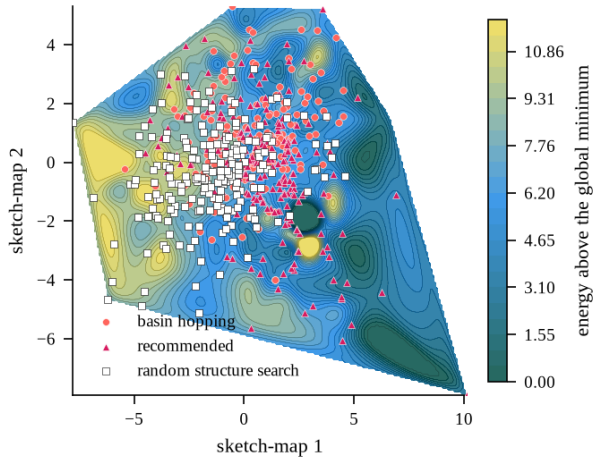


Figure S4: LJ_{38} landscape on that sketch-map. Colour is the energy above the global minimum. Basin hopping sits 2.429 above it. Random structure search sits 3.051 above it.

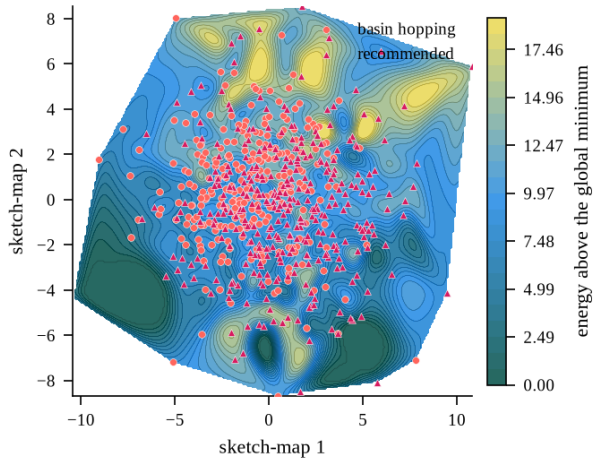


Figure S5: LJ_{75} landscape on its sketch-map. Colour is the energy above the Marks decahedron. Basin hopping and the recommended hop both sit 1.210 above it.

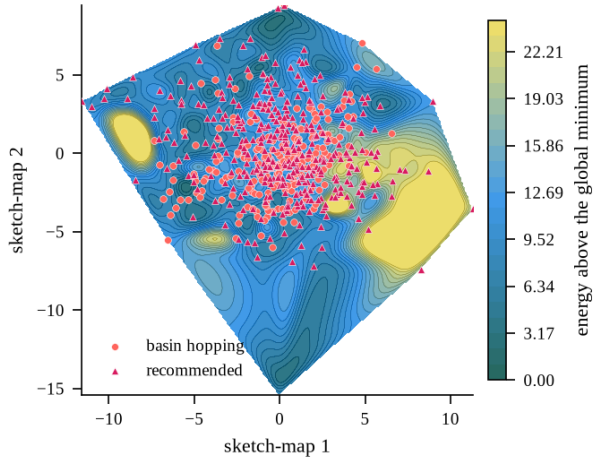


Figure S6: LJ_{98} landscape on its sketch-map. Colour is the energy above the Leary minimum. The recommended hop sits 0.022 above it. Basin hopping sits 0.119 above it.

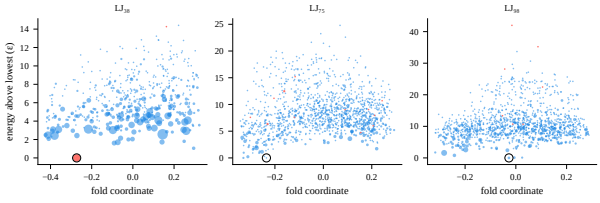


Figure S7: Occupancy folds of LJ_{38}, LJ_{75} and LJ_{98}. Each panel is energy above the lowest minimum. Against the fold coordinate.

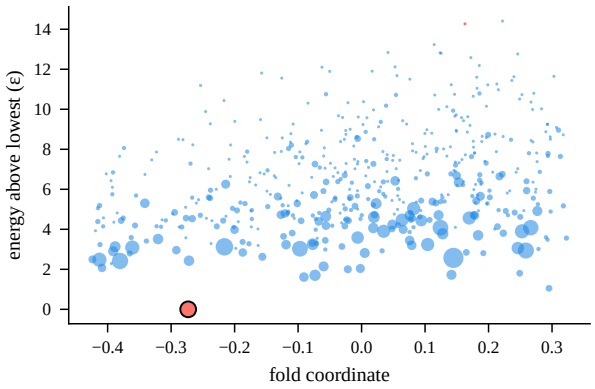


Figure S8: LJ_{38} occupancy fold. The arms are not marked. In the sealed table, the recommended hop reaches the fcc minimum. And basin hopping stops 2.429 above it.

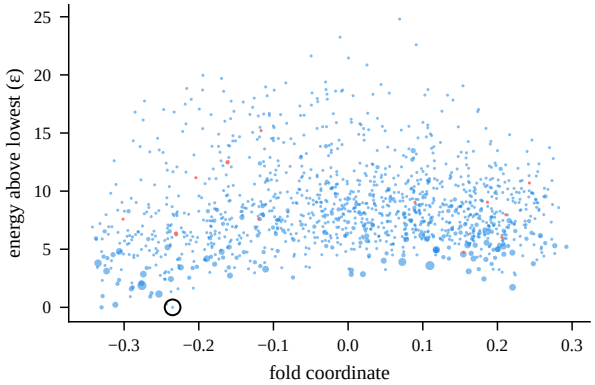


Figure S9: LJ_{75} occupancy fold. The arms are not marked. In the sealed table. Both arms stop 1.210 above the Marks decahedron.

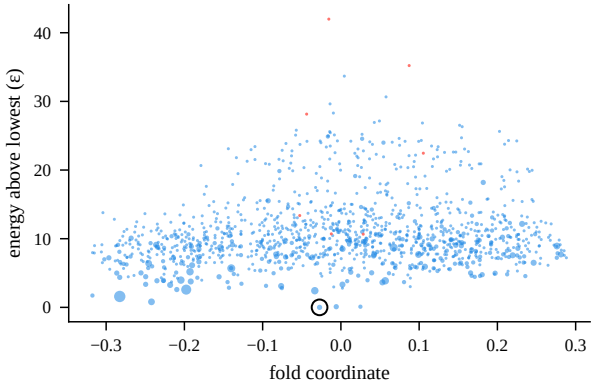


Figure S10: LJ_{98} occupancy fold. The arms are not marked. In the sealed table. The recommended hop stops 0.022 above the Leary minimum. And basin hopping 0.119 above it.

Comparison of simple potential functions for simulating liquid water. *J. Chem. Phys.* **1983**, *79*, 926–935.

(S4) Dieterich, J. M.; Hartke, B. OGOLEM: Global cluster structure optimisation for arbitrary mixtures of flexible molecules. A multiscaling, object-oriented approach. *Mol. Phys.* **2010**, *108*, 279–291.

(S5) Zhang, J.; Dolg, M. ABCluster: the artificial bee colony algorithm for cluster global optimization. *Phys. Chem. Chem. Phys.* **2015**, *17*, 24173–24181.